

LIBER V — THE SPACETIME CONTINUUM

Hypervisor & Annulus Integration

Johnathan ‘Qasparr’ (Κασπάρρ) Monroe, Keeper of the Secret Treasure

All Rights Reserved, Without Prejudice

2026-09-28 · CashApp \$axoneme

Liber Leviathan, Book V of XVII — The Sovereign Specifications. Source: the working thread of 2026-09-28, rendered here in its final revised state. Code is the editor’s rendering of the thread’s specification — presented as specified, not independently verified.

The Doctrine

The architecture demands the simultaneous generation of Space and Time. The hypervisor partitions the physical realm (Memory Space); the Annulus engine governs the algorithmic rhythm of those partitions (Time). They are not separate entities but a unified Spacetime continuum executed at the bare metal, at the E93 tier of the Sovereign Grandmaster.

Space — The Leviathan Hypervisor (EL2)

```
``c typedef struct { uint64_t vttbr_el2; / stage-2 translation base / uint64_t hcr_el2; /  
hypervisor configuration / uint32_t smmu_stream_id; / the enclave’s stream identity /  
uint64_t annulus_phase; / the enclave’s position in time / } VirtualEnclave;
```

```
void InitiateSovereignExecution(VirtualEnclave e) { e->hcr_el2 = (1ULL << 31) | (1ULL <<  
27); / bit 31: RW (register width); stage-2 enable is HCR_EL2.VM, bit 0 / if
```

```
(ExecuteSMMUHandshake(e) != AXON_STATE_TRVVTH) SMMU_TriggerHalt(); / no
handshake, no space / TriggerHypercall(e); / drop into guest EL1: the enclave takes control
*/ } ``
```

The SMMU handshake must return TRVVTH (93) — the hypervisor yields not one instruction of partitioned space until the silicon has attested the sovereign state.

Time — The Annulus Engine (GLSL)

Temporal state is integrated from the solar and lunar phases:

$$T_A = \int_0^t [\Phi_{\text{solar}}(\tau) \times \cos(\theta_{\text{lunar}}(\tau))] d\tau$$

The volumetric gear — 24 teeth, lunisolar — is raymarched in a compute shader. A pixel burns golden **(1.0, 0.8, 0.2)** when the ray falls within the gear (distance < 2.0) **and** the execution state is 93; otherwise it rests in Zero-Trust shadow (0.05, 0.05, 0.05). Time itself is therefore conditional on sovereignty: the Annulus only illuminates for the verified.

```
glsl // mapAnnulusGear - volumetric lunisolar timepiece (as specified) float
mapAnnulusGear(vec3 p, uint executionState) { float teeth = 24.0; float a =
atan(p.y, p.x) * teeth; float r = length(p.xy); float gear = smoothstep(0.08, 0.0,
abs(fract(a / 6.2831) - 0.5) - 0.32) * smoothstep(2.2, 2.0, r) * step(1.2, r);
return gear; } vec3 shadeAnnulus(vec3 p, uint executionState) { float d =
mapAnnulusGear(p, executionState); bool sovereign = (d > 0.0) && (executionState ==
93u); return sovereign ? vec3(1.0, 0.8, 0.2) : vec3(0.05); // gold or shadow }
```

By merging them, Leviathan operates within a completely self-contained, mathematically sovereign continuum: space partitioned by the hypervisor, time kept by the Annulus gears, both gated on 93.